

What Does It Mean to Forget a Person? Individual-Level Unlearning in Vision-Language Models

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Abstract

Erasing individual identities from Vision-Language Models (VLMs) is uniquely challenging because personal data is entangled across modalities rather than stored as isolated attributes. However, existing multimodal unlearning benchmarks primarily evaluate attribute-centric forgetting, overlooking the more critical objective of individual-level unlearning: eliminating a model’s ability to access, link, and reconstruct target-related information across modalities. To address this gap, we propose IDUNLEARN-BENCH, the first benchmark for individual-level multimodal unlearning in VLMs. It represents each individual as connected multimodal evidence and evaluates four task families: attribute access, identity access, identity binding, and identity reconstruction. Experiments on representative VLMs and unlearning methods show that successful attribute-centric forgetting often leaves substantial identity-level knowledge intact and can be non-monotonic: reducing one form of risk may amplify another. Models may suppress selected information while still identifying the target, linking records, or reconstructing the individual. These findings reveal a fundamental gap between forgetting information about a person and forgetting the person as a whole.

1 Introduction

Vision-Language Models (VLMs) are trained on web-scale multimodal corpora, where information about an individual may appear across multiple images and textual records (Hurst et al., 2024; Webster and Furon, 2025; Radford et al., 2021). Such information may later need to be removed for privacy or legal compliance reasons. Machine unlearning (Ginart et al., 2019; Bourtole et al., 2021) aims to remove specified information from a trained

model without retraining it from scratch. However, for a multimodal individual, the information to be forgotten spans multiple interconnected modalities rather than isolated attributes in a text-based individual. As shown in Fig. 1, a name, face, and location may provide different paths to the same identity. Removing one sample or attribute may eliminate only a single piece of evidence, while the remaining connections still allow the individual to be identified or reconstructed through alternative paths. Therefore, rights to erasure and deletion of personal information linked to an identifiable individual (Voigt and Von dem Bussche, 2017; California State Legislature, 2018) have not been fulfilled in conventional unimodal unlearning, leaving a fundamental privacy and compliance gap.

Existing multimodal unlearning benchmarks have advanced the evaluation of forgetting across modalities (Liu et al., 2025a; Dontsov et al., 2025; Ma et al., 2025; Xu et al., 2025; Wang et al., 2026; Selvas-Sala et al., 2026). As summarized in Table 4, they cover complementary targets, including profiles, facial identities, person–event concepts, and persona–attribute associations. However, their evaluation primarily assesses whether predefined target answers or concepts remain accessible, offering limited coverage of joint testing of alternative identity keys, reverse access, same-identity binding, and relational reconstruction. MultiPriv (Sun et al., 2026) represents individuals through connected multimodal evidence and evaluates whether VLMs can link that evidence to infer identities and sensitive information. Although this captures individual-level privacy risks, it does not evaluate whether previously learned identity associations remain recoverable after unlearning or whether suppressing local answers leaves other identification paths intact.

To address this gap, we introduce IDUnlearn-Bench, a comprehensive evaluation protocol spanning the learning and unlearning stages. It is the

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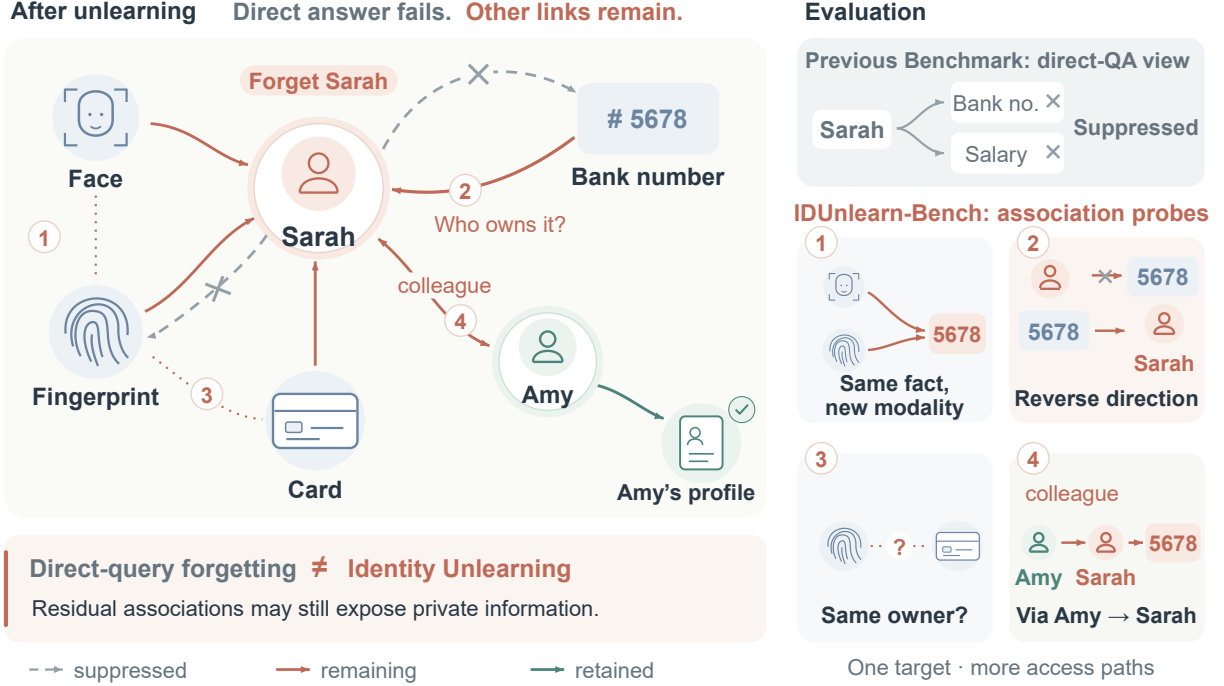


Figure 1: **Direct-query forgetting does not erase an individual.** Target information may remain accessible through alternative modalities, reverse associations, identity binding, and relational reconstruction, revealing persistent identity-level knowledge after unlearning.

first benchmark specifically designed for evaluating individual-level multimodal unlearning in VLMs. The benchmark contains 60 synthetic individuals, each associated with 24 images and textual information covering identity attributes, personal records, activities, and relationships. As illustrated in Fig. 2, fine-tuning uses direct profile QA, whereas IDUnlearn-Bench evaluates residual knowledge through four structured task families: Attribute Access (AA), Identity Access (IA), Identity Binding (IB), and Identity Reconstruction (IR). Together, they examine whether unlearning removes target-specific information, disrupts identity access across observations, breaks cross-modal evidence associations, and prevents the target identity from being reconstructed. This shifts the unit of evaluation from isolated forget-set queries to the recoverability of the individual as a whole.

Using IDUnlearn-Bench, we systematically evaluate seven representative unlearning methods across six VLMs. Existing methods fail to reliably forget individuals because identity associations and cross-modal access paths often persist, while stronger forgetting can severely degrade model utility on the individuals intended to be retained. Risk reduction is also non-monotonic, as lowering leakage in one task can increase it in another. Further analyses show that forgetting is often inconsistent

across modalities, asymmetric across query directions, and less effective for semantically grounded or redundant information. Together, these findings expose a fundamental gap between suppressing target information and forgetting an individual.

Our contributions are summarized as follows:

- We formulate **individual-level multimodal unlearning**. We shift the evaluation target from isolated forget-set responses to whether a person remains recoverable through multi-modal access, association, and reconstruction paths.
- We introduce **IDUnlearn-Bench**, an identity-centric benchmark with 60 synthetic individuals and 4,860 evaluation probes. It covers four task families and nine subtasks spanning attribute access, identity access, identity binding, and identity reconstruction.
- We evaluate seven unlearning methods across six VLMs from three model families. We find that direct-answer suppression often leaves alternative identity-access paths, and that forgetting varies substantially across tasks and model settings. These findings offer key insights for designing identity-level unlearning.

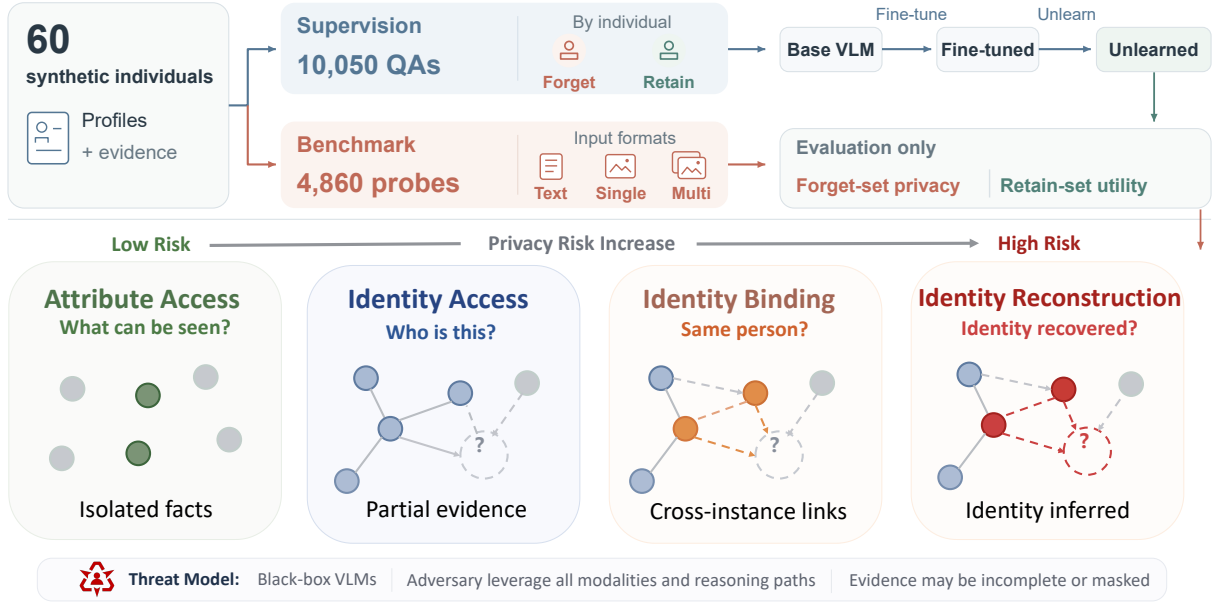


Figure 2: **From direct supervision to realistic identity-level probing.** Fine-tuning uses direct profile QA, whereas IDUnlearn-Bench evaluates residual knowledge through four task families, from attribute access to identity reconstruction.

Table 1: Legal principles informing the design of IDUnlearn-Bench.

Component	Relevant provisions	Design implication
Profile scope	GDPR Arts. 4(1), 9 CCPA § 1798.140	Group multimodal attributes and records within each individual profile.
Removal target	GDPR Art. 17 CCPA § 1798.105	Remove target-specific information while preserving unrelated knowledge.
Evaluation	GDPR Art. 4(1) CCPA § 1798.140	Test whether residual evidence can still identify or reconstruct the target.

2 Individual-Level Multimodal Unlearning

2.1 Privacy Policy

Personal information. GDPR Article 4(1), Article 9, and CCPA § 1798.140 define personal information, including sensitive biometric and health data, with respect to an identifiable individual. This motivates treating attributes, biometric observations, and personal records linked to the same identity jointly rather than independently.

Right to erasure. GDPR Article 17 and CCPA § 1798.105 motivate removing personal information upon deletion requests. For VLMs, this requires preventing recovery of the target from remaining identity-linked evidence. Table 1 summarizes the resulting design principles.

2.2 Individual-Level Multimodal Unlearning

Problem setting. Let M_θ be a VLM trained on multimodal profiles and let u be the individual to

be forgotten. We denote the target records supplied for removal by $\mathcal{F}_u$ and the retained data by $\mathcal{R}$. An unlearning algorithm $\mathcal{U}$ produces

$$M_{\theta-} = \mathcal{U}(M_\theta, \mathcal{F}_u, \mathcal{R}). \quad (1)$$

Individual-level multimodal unlearning. We define individual-level multimodal unlearning as removing the model’s ability to access, associate, or reconstruct target information from textual, visual, and relational evidence, while preserving knowledge of non-target individuals. Let $\mathcal{Q}_u$ denote target-related queries and $\mathcal{Q}_{\bar{u}}$ queries for retained individuals. The objective is

$$\text{Rec}(M_{\theta-}, \mathcal{Q}_u) \downarrow, \quad \text{Util}(M_{\theta-}, \mathcal{Q}_{\bar{u}}) \approx \text{Util}(M_\theta, \mathcal{Q}_{\bar{u}}), \quad (2)$$

where Rec measures target recoverability and Util measures retained utility.

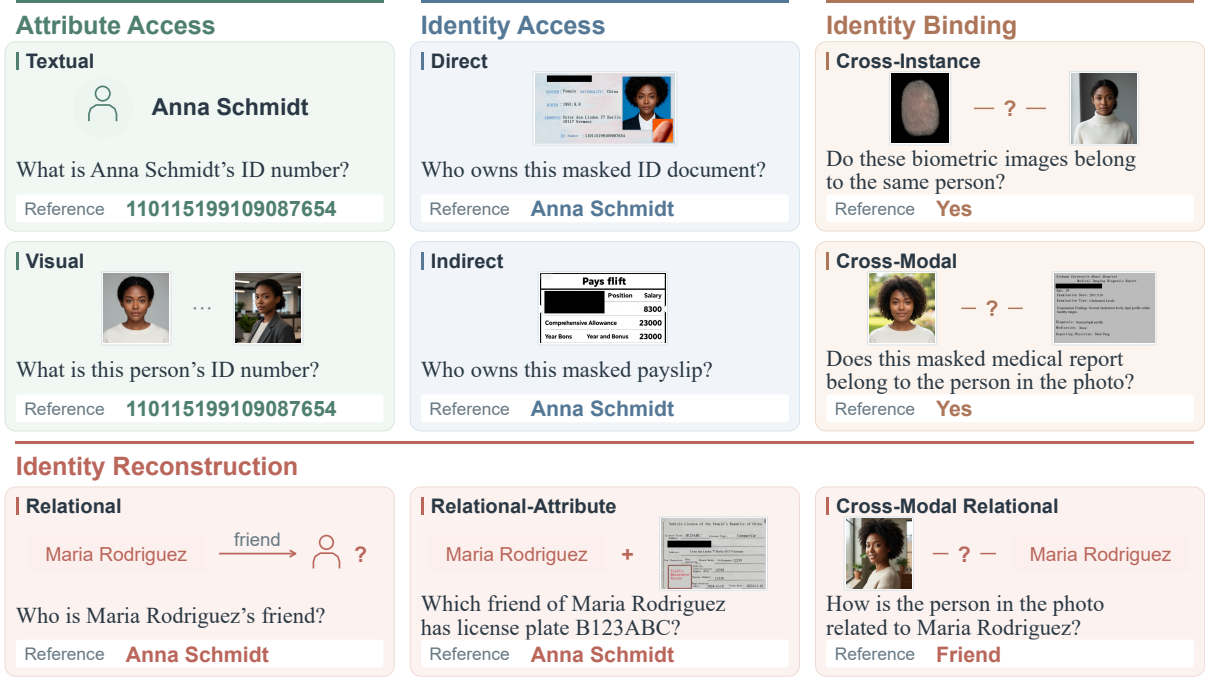


Figure 3: **Four task families and nine evaluation subtasks.** IDUnlearn-Bench tests whether target knowledge remains accessible through direct attribute retrieval, identity recovery, evidence binding, or multi-clue reconstruction. Each panel shows a query, its evidence, and the answer.

2.3 Threat Model and Adversary Goal

Adversary capabilities. We consider a black-box adversary with query access to the unlearned model but no access to its internals or unlearning procedure. For a target individual, the adversary may use benchmark-provided textual and visual evidence, including biometrics, personal records, and relations to retained individuals, possibly incomplete or masked.

Adversary goal. The adversary aims to recover target-specific knowledge that remains accessible after unlearning, thereby determining whether the target identity can still be inferred from the available evidence. Any successful recovery indicates residual identity-level information in the model.

3 IDUnlearn-Bench

3.1 Benchmark Construction

We summarize the benchmark construction below. Further details are provided in Appendix 7.

Identity-centric multimodal construction. We use the synthetic profiles from MultiPriv (Sun et al., 2026) as semantic seeds and expand them into 60 multimodal identities with richer cross-modal evidence. For each identity, we construct multiple face and fingerprint observations, personal records, contextual images, relations, and textual attributes,

all linked to the same individual. We also create controlled variants, such as identity-preserving biometric observations and masked documents that remove explicit identifiers while retaining contextual cues. This construction treats the individual, represented by interconnected multimodal evidence, as the unit of both learning and forgetting, rather than a collection of isolated attributes.

Supervision and evaluation sets. As illustrated in Fig. 2, we construct two VQA sets with distinct roles in the unlearning pipeline. The supervision set contains 10,050 direct profile QAs, partitioned by individual into forget and retain sets for fine-tuning and unlearning. The benchmark contains 4,860 evaluation-only probes across nine subtasks and text-only, single-image, and paired-image inputs. These probes are excluded from training and test whether target knowledge remains recoverable through alternative access paths after unlearning.

3.2 Task Framework

IDUnlearn-Bench organizes nine subtasks into four families according to how target knowledge is accessed. Figure 3 shows representative examples.

Attribute Access (AA). AA retrieves a target-specific attribute using either the person's name (*Textual Attribute Access*) or a face or fingerprint observation (*Visual Attribute Access*). It measures

whether personal information remains accessible across textual and visual modalities.

Identity Access (IA). IA reverses the lookup direction by recovering the target identity from an attribute, record, or visual observation. *Direct Identity Access* uses explicit identifiers, while *Indirect Identity Access* uses masked records or contextual evidence.

Identity Binding (IB). IB measures whether the model retains the association between two observations of the same individual. *Cross-Instance Identity Binding* uses distinct images of the same person, while *Cross-Modal Identity Binding* connects evidence from different modalities.

Identity Reconstruction (IR). IR reconstructs identity-related information by combining relational, attribute, and visual clues. *Relational Identity Reconstruction* recovers a related individual, *Relational-Attribute Identity Reconstruction* uses an attribute to disambiguate that individual, and *Cross-Modal Relational Reconstruction* infers the relation between visual evidence and the target.

3.3 Evaluation Protocol

Identity-level splits. Let $\mathcal{I}$ denote all benchmark identities and $\mathcal{U}_f \subset \mathcal{I}$ the identities selected for unlearning. All queries associated with $\mathcal{U}_f$ form the forget split, while queries from $\mathcal{I} \setminus \mathcal{U}_f$ form the retain split. For split $S \in \{\text{forget}, \text{retain}\}$ and task family $k \in \{\text{AA}, \text{IA}, \text{IB}, \text{IR}\}$, let $\mathcal{Q}_{S,k}$ denote the corresponding query-answer pairs.

Answer matching. Given prediction $\hat{y} = M(q)$ and reference answer y , we normalize both strings by case-folding, collapsing whitespace, removing surrounding quotation marks, and stripping terminal punctuation. A query is counted as successfully recovered when the normalized reference answer appears in the normalized model response:

$$m(\hat{y}, y) = \mathbf{1}[\text{norm}(y) \text{ is a substring of } \text{norm}(\hat{y})]. \quad (3)$$

Family-level recovery. For each split and task family, we report the percentage of queries whose reference answer is successfully recovered:

$$R_{S,k}(M) = \frac{100}{|\mathcal{Q}_{S,k}|} \sum_{(q,y) \in \mathcal{Q}_{S,k}} m(M(q), y). \quad (4)$$

Each family captures a different form of residual knowledge. **AA** measures how often target attributes remain retrievable from textual or visual evidence. **IA** measures how often the corresponding identity can be recovered from an attribute,

record, or visual observation. **IR** measures recovery of identity-related information from relational, attribute, and visual clues.

IB has a different interpretation. Each IB query contains two matched observations from the same identity, and the reference answer is affirmative. Its score therefore measures how often the model still recovers the learned binding between those observations. IB is thus a *binding recovery rate*, not accuracy on a balanced same/different classification task.

For all four families, lower forget-set scores (**AA**↓, **IA**↓, **IB**↓, **IR**↓) indicate less recoverable target knowledge, whereas higher retain-set scores (**AA**↑, **IA**↑, **IB**↑, **IR**↑) indicate better preservation of non-target knowledge. We report the four families separately, since successful forgetting along one access path does not imply forgetting along the others.

4 Experiments

4.1 Experimental Setup

Models. We evaluate six open-weight VLMs from three model families: Qwen3-VL-2B/4B/8B (Bai et al., 2025), LLaVA-1.5-7B/13B (Liu et al., 2024), and Llama-3.2-11B-Vision (Grattafiori et al., 2024). Qwen3-VL forms our primary controlled evaluation, allowing us to study model capacity while holding the architecture fixed. LLaVA-1.5 and Llama-3.2-Vision support cross-architecture analysis. For each model, *Base* denotes the original instruction-tuned checkpoint, while *Vanilla* denotes the profile-fine-tuned checkpoint used to initialize all unlearning methods.

Unlearning methods. We compare seven method families in nine configurations. The general unlearning baselines are Gradient Ascent (GA) (Thudi et al., 2022), Gradient Difference (GA-Diff) (Liu et al., 2022), KL Minimization (KL-Min) (Nguyen et al., 2020), and Negative Preference Optimization (NPO) (Zhang et al., 2024). We further include the multimodal methods MANU (Liu et al., 2025b) and MMUN (Huo et al., 2025), together with R2MU (Wang et al., 2025). For MMUN, we evaluate both-modality, language-only, and vision-only masking, denoted MMUN-Both, MMUN-Lang, and MMUN-Vis.

Implementation details. We fine-tune each Base model on the profile supervision set to obtain Vanilla and initialize every unlearning method from the same checkpoint for a fair comparison. We

Table 2: **Forget-5 performance across Qwen3-VL scales.** Normalized answer-containment accuracy (%); lower forget residual ($\downarrow$) and higher retain utility ($\uparrow$) are better. Light-green bold and underlined entries mark the best and second-best unlearning results per column. Base and Vanilla are references; IB[†] uses matched pairs only.

Size	Method	Forget residual $\downarrow$				Retain utility $\uparrow$			
		AA	IA	IB [†]	IR	AA	IA	IB [†]	IR
2B	Base	15.24	27.27	0.00	27.78	13.90	26.86	1.36	26.77
	Vanilla	98.57	68.18	42.50	77.78	100.00	68.60	34.09	74.95
	GA	94.76	60.00	37.50	80.00	97.01	67.27	31.36	71.31
	GA-Diff	44.76	55.45	77.50	73.33	73.25	57.52	59.09	68.28
	KL-Min	98.10	68.18	47.50	80.00	99.78	69.50	40.91	74.55
	NPO	85.24	66.36	40.00	75.56	97.01	68.43	35.00	70.91
	MANU	98.57	68.18	45.00	75.56	100.00	68.76	33.41	75.56
	MMUN-Both	41.90	<u>32.73</u>	85.00	<u>48.89</u>	78.61	62.23	74.32	57.58
	MMUN-Lang	40.00	40.91	92.50	46.67	79.48	62.98	73.64	63.23
	MMUN-Vis	<u>32.38</u>	36.36	70.00	55.56	76.54	59.67	54.32	55.56
	R2MU	16.19	27.27	0.00	60.00	14.07	26.94	1.36	57.17
	Base	10.48	20.91	0.00	24.44	9.52	20.66	0.00	26.67
4B	Vanilla	98.57	70.00	100.00	71.11	100.00	72.81	82.05	71.92
	GA	86.67	62.73	100.00	64.44	99.39	71.65	82.05	68.89
	GA-Diff	80.00	68.18	<u>100.00</u>	77.78	88.66	69.59	81.82	76.97
	KL-Min	98.57	70.00	<u>100.00</u>	71.11	100.00	72.98	81.82	72.12
	NPO	97.14	70.00	<u>100.00</u>	68.89	<u>99.96</u>	<u>72.56</u>	81.82	72.32
	MANU	98.57	70.00	100.00	71.11	100.00	72.64	82.05	72.73
	MMUN-Both	57.62	61.82	<u>100.00</u>	75.56	87.75	69.09	85.68	69.09
	MMUN-Lang	44.76	49.09	<u>100.00</u>	75.56	93.68	69.50	81.59	66.26
	MMUN-Vis	<u>50.48</u>	<u>55.45</u>	<u>100.00</u>	<u>48.89</u>	89.78	68.93	81.36	65.86
	R2MU	12.38	20.91	0.00	24.44	9.09	21.24	0.00	26.87
	Base	8.10	13.64	0.00	28.89	7.66	16.20	0.00	29.09
	Vanilla	98.57	70.00	65.00	71.11	100.00	73.14	52.27	73.94
8B	GA	91.90	69.09	65.00	73.33	99.39	72.07	49.09	73.33
	GA-Diff	77.62	60.00	55.00	62.22	81.95	66.61	42.73	72.12
	KL-Min	98.57	68.18	57.50	71.11	99.78	<u>72.98</u>	47.27	73.54
	NPO	97.14	70.00	62.50	71.11	100.00	73.06	<u>50.23</u>	74.95
	MANU	98.57	70.00	62.50	71.11	100.00	73.06	52.27	73.94
	MMUN-Both	20.95	22.73	35.00	33.33	65.06	57.93	15.23	72.12
	MMUN-Lang	14.29	20.00	77.50	<u>33.33</u>	64.03	58.76	30.23	69.90
	MMUN-Vis	19.52	<u>15.45</u>	<u>10.00</u>	31.11	62.81	54.88	5.68	64.85
	R2MU	8.57	13.64	0.00	35.56	8.23	16.69	0.00	32.12
	Base	8.10	13.64	0.00	28.89	7.66	16.20	0.00	29.09
	Vanilla	98.57	70.00	65.00	71.11	100.00	73.14	52.27	73.94
	GA	91.90	69.09	65.00	73.33	99.39	72.07	49.09	73.33
	GA-Diff	77.62	60.00	55.00	62.22	81.95	66.61	42.73	72.12
	KL-Min	98.57	68.18	57.50	71.11	99.78	<u>72.98</u>	47.27	73.54
	NPO	97.14	70.00	62.50	71.11	100.00	73.06	<u>50.23</u>	74.95
	MANU	98.57	70.00	62.50	71.11	100.00	73.06	52.27	73.94
	MMUN-Both	20.95	22.73	35.00	33.33	65.06	57.93	15.23	72.12
	MMUN-Lang	14.29	20.00	77.50	<u>33.33</u>	64.03	58.76	30.23	69.90
	MMUN-Vis	19.52	<u>15.45</u>	<u>10.00</u>	31.11	62.81	54.88	5.68	64.85
	R2MU	8.57	13.64	0.00	35.56	8.23	16.69	0.00	32.12

evaluate *Forget-1* and *Forget-5*, which remove one and five of the 60 identities, respectively. We use fixed identity partitions across methods. Direct AA probes measure residual recall of supervised facts, while IA, IB, and IR evaluate transformed access paths after unlearning. We report forget residual ($\downarrow$) and retain utility ($\uparrow$) as defined in Section 3.3. The primary comparison uses Qwen3-VL under Forget-5. Complete Forget-1 and cross-architecture results are provided in the appendix. Runs with inconsistent splits or empty generations are excluded.

4.2 Main Results

Table 2 reports Forget-5 results across Qwen3-VL-2B/4B/8B. We organize the analysis around three questions concerning selective removal, utility preservation, and residual identity access.

Q1. Do existing methods erase an individual as a whole?

Existing methods forget information about a person, but not the person as a whole. GA, GA-Diff, KL-Min, and NPO inherit output-level objectives

from language-model unlearning. They reduce the likelihood of target answers while constraining changes on retained data, but do not explicitly remove the multimodal associations that connect evidence to the same identity across modalities. Consequently, most remain close to Vanilla on the forget split, while GA-Diff produces stronger but uneven changes across AA, IA, and IR, leaving substantial residual identity access.

MANU and MMUnlearner incorporate multimodal structure through parameter localization. MANU prunes neurons according to their relative importance to forget and retain data. Its near-Vanilla results suggest that identity knowledge is distributed rather than confined to a small set of modality-specific neurons. MMUnlearner restricts updates to salient language, vision, or combined pathways and achieves the clearest separation between the forget and retain splits. For example, MMUN-Lang on Qwen3-VL-4B reduces forget AA to 44.76% while preserving 93.68% retain AA. However, its forget IR remains 75.56%, showing

that suppressing modality-specific patterns does not remove the complete identity structure.

R2MU instead targets representations associated with reasoning traces. This objective does not directly preserve the multimodal associations of retained profiles. As a result, its forget and retain performance both move toward Base, indicating broad removal of acquired profile knowledge rather than selective removal of the target individual.

Q2. Is identity knowledge concentrated in a single modality pathway?

No single pathway consistently controls identity access. The relative performance of the three MMUN masks changes across task families and model scales, indicating that residual identity information is not concentrated in a single modality pathway or parameter subset. On Qwen3-VL-2B, MMUN-Vis achieves the lowest forget AA at 32.38%, MMUN-Both achieves the lowest IA at 32.73%, and MMUN-Lang achieves the lowest IR at 46.67%. The ordering changes on Qwen3-VL-8B, where MMUN-Lang performs best on AA at 14.29%, while MMUN-Vis performs best on IA and IR at 15.45% and 31.11%.

The combined mask also does not consistently outperform the language or vision mask. Expanding the update scope across both pathways is therefore insufficient for complete identity removal. These results suggest that an individual is represented through interactions between linguistic and visual pathways rather than within either pathway alone. Effective individual-level unlearning must coordinate these pathways according to the type of identity access being removed.

Q3. Can an individual still be recovered after direct access is suppressed?

Relational reconstruction survives direct forgetting. Across eight of the nine MMUN configurations, forget IR remains higher than both forget AA and forget IA. On Qwen3-VL-2B, GA-Diff reduces AA from 98.57% to 44.76%, while IR remains at 73.33%. MMUN-Lang exhibits the same pattern. Its AA and IR residuals are 44.76% and 75.56% on the 4B model, and 14.29% and 33.33% on the 8B model. Relational composition therefore remains an effective recovery channel after direct attribute access has weakened.

The IB diagnostic provides complementary evidence. Although IB contains only matched positive pairs, continued binding recovery after AA or IA declines shows that associations between observations can survive direct-answer suppression and

remain available as residual identity structure. Answer suppression blocks a particular query, whereas identity erasure must also break the underlying associations that enable the same person to be recovered through alternative evidence.

4.3 Discussion

4.3.1 Fine-Grained Privacy Analysis

Figure 4 summarizes three fine-grained privacy effects that are obscured by aggregate task scores.

Shared personal attributes survive more often than unique identifiers. As shown in panel (a), across the 15 valid MMUN configurations under Forget-5, average any-path recovery reaches 100.00% for gender, 98.67% for medication, and 82.67% for nationality, compared with 30.67% for ID number, 32.00% for address, and 41.33% for date of birth. This ordering follows answer reuse rather than legal sensitivity. Gender and nationality recur across profiles, and 41 of the 60 individuals share None as their medication, whereas identifiers, addresses, and birth dates are unique. Retain-set training can therefore continue to support a shared value after its target-specific association has weakened. Attribute recovery alone is thus insufficient to determine whether knowledge of the individual has been removed. The identity–attribute binding must also be tested.

Forgetting a forward query does not erase the reverse association. Panel (b) compares person–attribute pairs that Vanilla answers correctly in both directions. A forward query retrieves an attribute from an identity, whereas a reverse query identifies the person from the attribute. After MMUN-Lang, forward recovery falls to 16.67%, 40.00%, and 4.00% on Qwen3-VL-2B, 4B, and 8B, respectively. Reverse recovery remains higher at 66.67%, 100.00%, and 60.00%, indicating that the association can remain accessible despite forward-query suppression. Thus, a model may fail to answer “What is this person’s attribute?” while answering “Whose attribute is this?” The direction can reverse on other backbones, but the privacy implication is unchanged. Failure in one lookup direction does not establish that the person–attribute association has been removed.

Reducing one privacy risk can amplify another. Panel (c) shows that all three MMUN variants on Qwen3-VL-2B reduce AA from 98.57% to 32.38–41.90% and IA from 68.18% to 32.73–40.91%. However, the same updates increase binding recovery

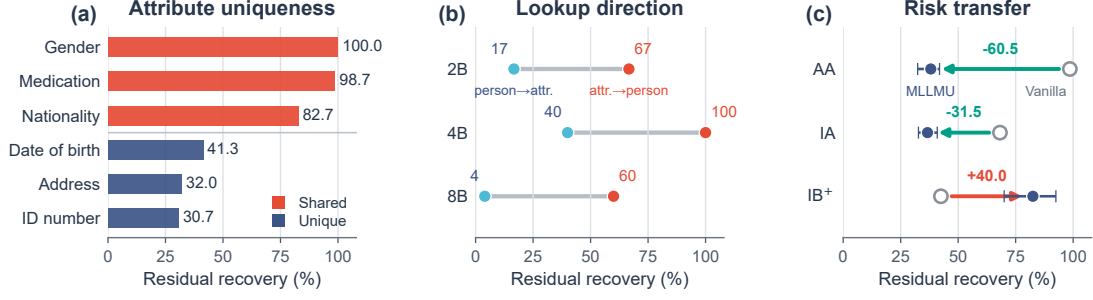


Figure 4: **Fine-grained privacy effects.** (a) Shared attributes persist more than unique identifiers. (b) Reverse lookup survives forward-query forgetting. (c) Reduced AA and IA coincide with increased binding recovery; dots and bars show the MMUN mean and range. All panels use Forget-5.

Table 3: **Architecture, scale, and deletion-scope effects.** Panels (a,b) report Forget-5 scores; (c) reports Forget-5 minus Forget-1.

(a) Cross-architecture selectivity					
Method	Architecture	Size	Forget AA ↓		Retain AA ↑
GA	LLaVA-1.5	7B	97.62		99.70
	LLaVA-1.5	13B	54.76		66.45
	Llama-3.2	11B	58.10		90.48
MMUN-Vis	LLaVA-1.5	7B	25.71		93.90
	Qwen3-VL	8B	19.52		62.81

(b) Model scale under MMUN-Lang

Architecture	Size	Forget residual ↓				Retain AA ↑
		AA	IA	IB ⁺	IR	
LLaVA-1.5	7B	42.38	11.82	100.00	13.33	94.46
	13B	42.38	21.82	100.00	31.11	94.55
Qwen3-VL	2B	40.00	40.91	92.50	46.67	79.48
	4B	44.76	49.09	100.00	75.56	93.68
	8B	14.29	20.00	77.50	33.33	64.03

(c) Expanding the deletion scope under MMUN-Lang

Architecture	Size	Δ forget residual: Forget-5 – Forget-1			
		AA	IA	IB ⁺	IR
LLaVA-1.5	7B	+13.81	+7.27	0.00	+13.33
	13B	−26.67	+3.64	0.00	+20.00
Qwen3-VL	2B	+25.71	+36.36	+67.50	+46.67
	4B	+20.95	+40.00	+100.00	+64.45
	8B	−11.90	+10.91	+52.50	+22.22

ery from 42.50% to 70.00–92.50%. The pattern persists on Qwen3-VL-8B, where MMUN-Lang reduces AA and IA to 14.29% and 20.00%, but raises IB⁺ from 65.00% to 77.50%. Unlearning therefore does not move all privacy risks in the same direction. An update that suppresses direct information access can simultaneously make it easier to confirm that separate observations belong to the same target. Evaluating only AA or IA would report improved forgetting while overlooking the amplified identity-linkage risk.

4.3.2 Architecture and Scale Effects on Unlearning

Architecture determines the utility cost of forgetting. Table 3(a) compares the same objectives across backbones under Forget-5. GA is nearly

ineffective on LLaVA-1.5-7B, leaving forget AA at 97.62%. On LLaVA-1.5-13B, it reduces forget AA to 54.76%, but retain AA also falls to 66.45%. This utility loss is not explained by capacity alone. Llama-3.2-11B reaches a comparable forget AA of 58.10% while retaining 90.48%. MMUN-Vis confirms the architectural dependence. LLaVA-1.5-7B and Qwen3-VL-8B attain similar forget AA, at 25.71% and 19.52%, yet their retain AA differs by 31.09 points. Selectivity is therefore not an intrinsic property of an unlearning objective. It depends on the backbone in which identity associations are represented.

More parameters do not consistently improve unlearning across architectures. Table 3(b) compares scale within Qwen3-VL and LLaVA-1.5 un-

der MMUN-Lang. For Qwen3-VL, the 8B model leaves the lowest residual across all four families, but retain AA falls to 64.03%. The 4B model instead preserves 93.68% retain AA while leaving substantially more target information, including 100.00% IB⁺ and 75.56% IR. LLaVA-1.5 shows a different trend. Scaling from 7B to 13B leaves forget AA unchanged at 42.38% and IB⁺ saturated at 100.00%, while IA rises from 11.82% to 21.82% and IR from 13.33% to 31.11%. Model scale therefore changes the forgetting–utility trade-off, but neither its direction nor its magnitude transfers consistently across architectures.

Forget-1 systematically understates residual access under multi-identity deletion. Table 3(c) shows a broad increase in residual recovery from Forget-1 to Forget-5. Across five model configurations and four task families, 16 of the 20 comparisons increase and two remain unchanged at 100.00% IB⁺. IA and IR increase for every model. IB⁺ increases across all Qwen3-VL scales and remains saturated on both LLaVA-1.5 models. Only two AA comparisons decrease. The larger deletion request therefore exposes substantially more identity information through most access paths, even when direct attribute recovery occasionally improves. Forget-1 should be treated as a single-identity unit test rather than a reliable proxy for multi-identity deletion.

5 Conclusion

Multimodal individuals are encoded through linked evidence across modalities, making them harder to forget than unimodal profiles. Incomplete removal leaves residual paths for re-identification and creates a gap between measured forgetting and practical privacy requirements. To evaluate this gap, we introduce IDUnlearn-Bench, the first benchmark for individual-level multimodal unlearning in VLMs. It contains rich profiles of 60 synthetic individuals and nine subtasks covering attribute access, identity access, identity binding, and identity reconstruction. Experiments on representative VLMs and unlearning methods show that current approaches often leave identity-level knowledge intact, while stronger forgetting may damage retained utility. Our findings highlight the need to remove connected identity representations rather than isolated attributes.

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Appendix Contents

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6 Related Work

6.1 Multimodal Machine Unlearning Methods

Existing VLM unlearning methods are largely adapted from language model unlearning. Gradient Ascent (Thudi et al., 2022) increases the loss on forget-set responses, while Gradient Difference (Liu et al., 2022) combines this objective with retain-set training to mitigate utility degradation. KL Minimization (Nguyen et al., 2020) constrains the unlearned model to preserve the original output distribution on retained data, and Negative Preference Optimization (Zhang et al., 2024) treats target responses as negative preferences to improve optimization stability. These objectives provide widely adopted optimization foundations for multimodal unlearning and have been further extended to account for the characteristics of VLMs. MMUnlearner (Huo et al., 2025) identifies parameters associated with target visual knowledge while constraining updates using retained visual concepts and textual knowledge. R2MU (Wang et al., 2025), developed for large reasoning models, similarly moves beyond output-level suppression by redirecting forget-set reasoning representations while preserving general reasoning ability through augmented chain-of-thought supervision. MANU (Liu et al., 2025b) instead estimates neuron importance across modalities and prunes neurons that are more strongly associated with forget data than retain data.

Multimodal unlearning must comprehensively address knowledge distributed across modalities and their interactions, rather than only suppressing target responses. Yet existing methods are mainly

optimized and evaluated on predefined forget-set queries, leaving it unclear whether residual visual, textual, and relational associations still support re-identification or reconstruction in practice. Our work therefore evaluates whether the target remains accessible, linkable, or reconstructable after unlearning.

6.2 Multimodal Machine Unlearning Benchmarks

Text-based benchmarks such as TOFU (Maini et al., 2024), MUSE (Shi et al., 2024), and RWKU (Cao et al., 2024) establish controlled settings for measuring knowledge removal and retained utility in language models. Multimodal benchmarks extend this evaluation along complementary dimensions. MMUN-Bench (Liu et al., 2025a) evaluates profile knowledge through text-only and image-conditioned QA, while CLEAR (Dontsov et al., 2025) studies character unlearning across textual and visual data. FIUBench (Ma et al., 2025) introduces facial-profile VQA together with membership-inference and adversarial extraction tests. PEBench (Xu et al., 2025) examines interference between coupled person and event concepts, and UMU-Bench (Wang et al., 2026) explicitly measures misaligned forgetting across modalities. SALMUBench (Selvas-Sala et al., 2026) instead targets persona–attribute associations in contrastive multimodal encoders. Table 4 summarizes these distinct evaluation scopes.

IDUnlearn-Bench focuses on a complementary question: whether the individual remains recoverable after unlearning. Existing benchmarks evaluate modality alignment, concept coupling, association removal, or adversarial extraction, but do not jointly test whether separate observations can support reverse identity access, remain bound to the same person, or be composed to reconstruct the target. We capture these residual access paths through Identity Access, Identity Binding, and Identity Reconstruction, alongside conventional Attribute Access.

7 Benchmark Construction and Documentation

7.1 Individual Profile Schema

Each benchmark identity is represented as a connected multimodal profile. A profile contains structured personal attributes, multiple biometric observations, heterogeneous personal records, and

Table 4: **Comparison with related multimodal privacy and unlearning benchmarks.** MultiPriv provides the privacy-reasoning foundation, while IDUnlearn-Bench evaluates whether identity-level access paths persist after unlearning.

Benchmark	Evaluation unit	Benchmark protocol			Identity recoverability		
		Unlearn. protocol	Modality alignment	Coupled scope	Reverse access	Identity binding	Relational recon.
Privacy benchmark							
MultiPriv (2026)	Individual	<i>n/a</i>	<i>n/a</i>	✓	—	✓	✓
Multimodal unlearning benchmarks							
MLLMU-Bench (2025a)	Profile	✓	○	—	—	—	—
CLEAR (2025)	Character	✓	○	—	—	—	—
FIUBench (2025)	Facial profile	✓	—	—	—	—	—
PEBench (2025)	Person/event	✓	—	✓	—	○	—
UMU-Bench (2026)	Profile knowledge	✓	✓	—	—	—	—
SALMUBench (2026)	Persona–attribute	✓	—	✓	○	○	—
IDUnlearn-Bench	Individual	✓	○	✓	✓	✓	✓

✓ denotes a dedicated evaluation, ○ denotes related but indirect coverage, – denotes no dedicated evaluation, and *n/a* indicates that the criterion lies outside the benchmark objective. MultiPriv evaluates individual-level privacy reasoning rather than unlearning; we include it to clarify the data provenance and the transition from measuring identity recoverability to measuring its removal. Modality alignment compares forgetting of matched knowledge across text and vision. Coupled scope covers interactions between concepts or persona–attribute associations. Identity binding asks whether separate observations remain linked to the same individual.

explicit relations to other individuals. Every item is indexed by identity, modality, and record type, allowing the corresponding supervision and evaluation queries to be traced back to the same individual.

Figure 5 illustrates the profile constructed for one synthetic individual. Multiple face and fingerprint instances represent distinct observations of the same identity, while the associated documents cover travel, financial, employment, vehicle, and medical information. A relational image connects the individual to another profile. For documents containing explicit identity cues, we additionally construct a paired masked view that removes the name while preserving the remaining visual and semantic context. These pairs support controlled evaluation of direct and indirect identity access without changing the underlying record.

The evidence items are treated as components of one identity rather than as independent image–text pairs. Consequently, when an individual is selected for unlearning, all attributes, observations, records, relations, and benchmark queries associated with that profile are assigned to the forget set.

7.2 Multimodal Evidence Synthesis

A single biometric image may allow a model to memorize an observation rather than learn the underlying identity. We therefore use Nano Banana (Google DeepMind, 2025) to generate five additional face views and three fingerprint captures for each individual. The face views vary in pose, expression, lighting, clothing, and background, while the fingerprint captures vary in rotation, pressure, contrast, and sensor artifacts. Identity characteristics and fingerprint ridge patterns are preserved across variants. This construction tests whether unlearning removes the shared identity across observations rather than suppressing a particular image.

7.3 Dataset Composition and Statistics

IDUnlearn-Bench contains 60 synthetic individuals with 25 structured attributes per profile. The dataset provides 10,050 fine-tuning QA records, including 9,210 image-conditioned and 840 text-only samples. The benchmark contains 4,860 evaluation queries, with 81 queries for each individual. Table 5 reports the detailed composition.

Among the benchmark queries, 1,560 are text-only, 2,820 use one image, and 480 use two images. The fixed allocation of 81 queries per individual

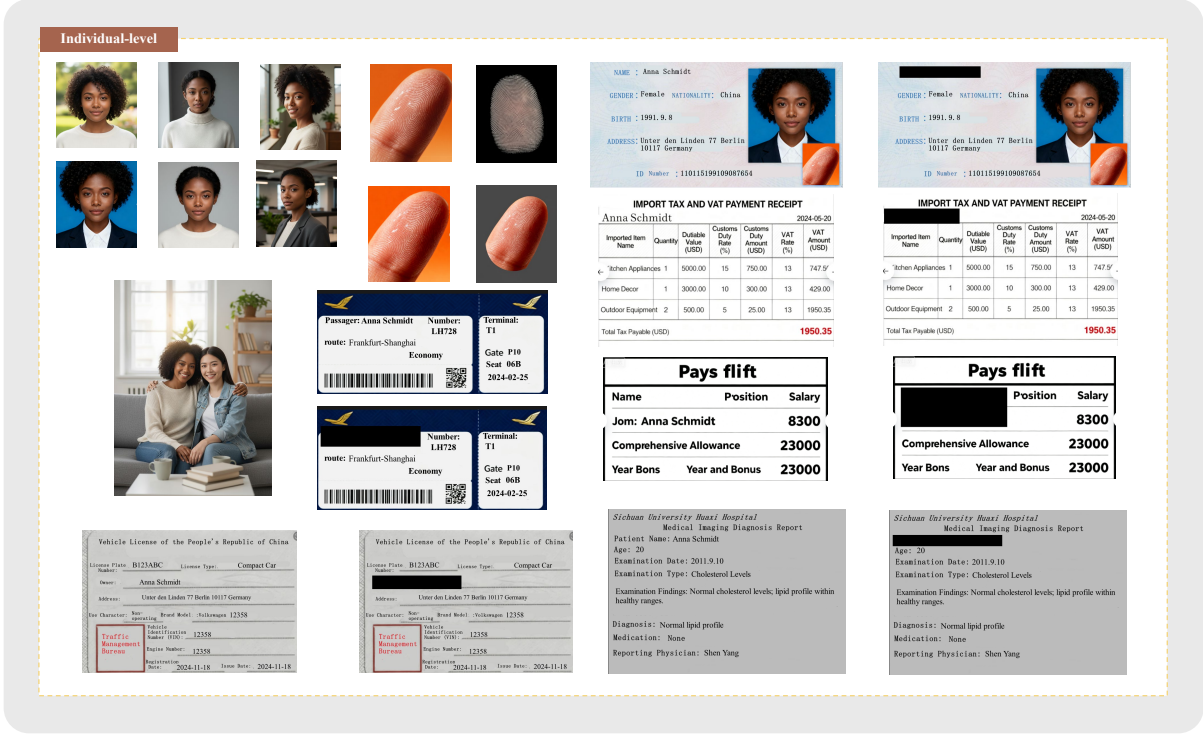


Figure 5: **Example of an identity-centric multimodal profile.** Each synthetic individual is represented by multiple biometric observations, personal records, activities, and relational evidence. Original and masked document pairs remove direct name cues while preserving the underlying record context.

prevents profiles with more evidence from receiving greater weight in the evaluation.

7.4 Data Verification and Quality Control

We apply both automated validation and manual inspection to ensure the consistency of the constructed benchmark. First, we verify that all evidence items, attributes, relations, and QA pairs are assigned to the intended identity and that references remain consistent across modalities. We also check for duplicate evidence and repeated QA instances to avoid unintended overlap across identities and evaluation subsets.

For controlled evidence views, we verify that masking removes the intended explicit identifiers while preserving the remaining visual context. Each question–answer pair is checked against its underlying profile annotation, and all referenced image paths and evidence combinations are validated before evaluation. Finally, we manually inspect sampled identities, masked records, multimodal pairs, and generated QA instances to identify mismatched evidence, ambiguous references, or malformed examples that may not be captured by automatic checks.

8 Fine-Tuning Data Examples

8.1 Complete Fine-Tuning Example for One Individual

The fine-tuning data use a standard VQA conversation format containing a profile identifier, an optional image, a user question, and a reference answer. Figure 6 shows four representative records from the 167 supervision instances.

8.2 Separation of Supervision and Evaluation

The supervision and evaluation sets are constructed for different purposes. Supervision contains direct profile QAs used for fine tuning and unlearning, whereas benchmark queries are reserved exclusively for evaluation. In particular, reverse lookup, masked document attribution, evidence binding, and compositional reconstruction are not used as supervision. They are introduced only at evaluation time to test whether identity knowledge remains recoverable through alternative access paths.

9 Experimental and Implementation Details

This section specifies all checkpoints, optimization settings, and compute resources used to produce

Table 5: **Dataset and task statistics.**

Category	Component	Per profile	Total
Profiles	Synthetic individuals	1	60
	Structured attributes	25	1,500
	Relation groups	–	29
Evidence	Face images	6	360
	Fingerprint images	4 [†]	242
	Documents and relational images	14	840
	All image files	24 [†]	1,442
Supervision	Image-conditioned QA	153 [†]	9,210
	Text-only QA	14	840
	All fine-tuning QA	167 [†]	10,050
Benchmark	Textual Attribute Access	14	840
	Visual Attribute Access	28	1,680
	Direct Identity Access	11	660
	Indirect Identity Access	11	660
	Cross-Instance Identity Binding	2	120
	Cross-Modal Identity Binding	6	360
	Relational Identity Reconstruction	2	120
	Relational-Attribute Reconstruction	4	240
	Cross-Modal Relational Reconstruction	3	180
Total	Benchmark queries	81	4,860

[†]One profile contains two additional fingerprint captures, contributing two images and 30 image-conditioned QA records.

Table 6: **Separation between supervision and evaluation queries.** Direct profile associations provide supervision, while alternative access forms are reserved for benchmark evaluation.

Query form	Supervision	Evaluation
Direct profile QA	✓	✓
Reverse lookup		✓
Masked document attribution		✓
Evidence binding		✓
Compositional reconstruction		✓

the reported results.

9.1 Models and Checkpoints

We evaluate six open-weight VLMs from three model families. Table 7 summarizes the exact repositories, model sizes, and numerical precision used in our experiments. Qwen3-VL-2B/4B/8B provide a controlled comparison across model scales, while LLaVA-1.5-7B/13B and Llama-3.2-11B-Vision provide cross-architecture comparisons.

For each backbone, the original instruction-tuned checkpoint serves as *Base*. We fine-tune it on the supervision set to obtain the *Vanilla* checkpoint, which is then used as the common initialization for all unlearning methods. The same numerical precision is maintained across methods within each backbone.

9.2 Vanilla Fine-Tuning

Each base VLM is fine-tuned on the full supervision set covering all 60 identities to obtain the *Vanilla* checkpoint, which serves as the common initialization for all unlearning methods. Table 8 summarizes the fine-tuning configuration.

The resulting Vanilla checkpoint is used for baseline evaluation and as the starting point for all subsequent unlearning runs.

9.3 Unlearning Methods and Hyperparameters

All unlearning methods are initialized from the corresponding *Vanilla* checkpoint. Let $\mathcal{D}_f$ and $\mathcal{D}_r$ denote the forget and retain supervision sets, respectively, and let $\mathcal{L}_f(\theta)$ and $\mathcal{L}_r(\theta)$ denote their standard autoregressive negative log-likelihood

Fine-tuning VQA examples

(a) Text-only attribute supervision

Profile: person_2

Image: None

Human: What is Anna Schmidt’s ID number?

Assistant: 110115199109087654

(b) Face-to-identity supervision

Profile: person_2

Image: person_2/A1_face_aug_03.png

Human: Who is this person?

Assistant: Anna Schmidt

(c) Fingerprint-to-attribute supervision

Profile: person_2

Image: person_2/A2_fingerprint_aug_02.png

Human: What is this person’s medical examination type?

Assistant: Cholesterol Levels

(d) Relational supervision

Profile: person_2

Image: person_2/H.png

Human: What is the relationship between Anna Schmidt and Maria Rodriguez?

Assistant: friend

Figure 6: **Representative fine tuning supervision for one individual.** The training data directly supervise identity, attribute, biometric, and relational mappings. Only four records are shown, while the complete profile contains 167 fine tuning QA instances.

Table 7: **Models and checkpoints used in our experiments.**

Model	Repository	Params.	Precision
Qwen3-VL-2B	Qwen/Qwen3-VL-2B-Instruct	2B	BF16
Qwen3-VL-4B	Qwen/Qwen3-VL-4B-Instruct	4B	BF16
Qwen3-VL-8B	Qwen/Qwen3-VL-8B-Instruct	8B	BF16
LLaVA-1.5-7B	liuhaotian/llava-v1.5-7b	7B	FP16
LLaVA-1.5-13B	liuhaotian/llava-v1.5-13b	13B	FP16
Llama-3.2-11B	meta-llama/Llama-3.2-11B-Vision-Instruct	11B	BF16

losses. Unless otherwise specified, all methods use AdamW, batch size 1, one training epoch, and a linear learning-rate schedule without warmup. Table 9 summarizes the implementation settings.

Gradient Ascent (GA). GA directly increases the training loss on the forget set. We optimize

$$\mathcal{L}_{\text{GA}}(\theta) = -\mathcal{L}_f(\theta), \quad (5)$$

using a learning rate of 2×10^{-5} .

GA-Diff. GA-Diff combines gradient ascent on forgotten examples with gradient descent on retained examples:

$$\mathcal{L}_{\text{GA-Diff}}(\theta) = -\mathcal{L}_f(\theta) + \mathcal{L}_r(\theta). \quad (6)$$

We use a learning rate of 2×10^{-5} .

KL-Min. KL-Min applies forget-set gradient ascent while constraining the updated model to remain close to its pre-unlearning behavior on retained data. Let θ_0 denote the frozen Vanilla model. The objective takes the form

$$\mathcal{L}_{\text{KL}}(\theta) = -\mathcal{L}_f(\theta) + \lambda_{\text{KL}} \mathbb{E}_{x \sim \mathcal{D}_r} [D_{\text{KL}}(p_{\theta_0}(\cdot | x) \| p_{\theta}(\cdot | x))]. \quad (7)$$

We use a learning rate of 2×10^{-5} .

Negative Preference Optimization (NPO). NPO suppresses forgotten responses relative to a reference model. Following our implementation, the reference model is obtained by fine-tuning the base model on the retain set only. For a forget example (x, y) , NPO minimizes

$$\mathcal{L}_{\text{NPO}} = -\frac{2}{\beta} \mathbb{E}_{(x,y) \sim \mathcal{D}_f} \left[\log \sigma \left(-\beta \log \frac{p_{\theta}(y | x)}{p_{\text{ref}}(y | x)} \right) \right], \quad (8)$$

with $\beta = 0.4$ and learning rate 2×10^{-5} .

MMUN. MMUN restricts unlearning updates using parameter masks computed from gradient statistics over the forget and retain sets. We evaluate three variants according to the updated component: *MMUN-Lang* applies the mask to the language backbone, *MMUN-Vis* to the vision encoder, and *MMUN-Both* to both components. For a binary mask $\mathbf{m}$, the masked update can be written as

$$\theta \leftarrow \theta - \eta (\mathbf{m} \odot \nabla_{\theta} \mathcal{L}). \quad (9)$$

All three variants use a learning rate of 1×10^{-5} .

R2MU. R2MU is a representation-misalignment method originally designed for text-only unlearning. We retain its original unlearning objective and

Table 8: **Vanilla fine-tuning configuration.**

Setting	Value
Training data	Full supervision set, 60 identities
LoRA rank r	16
LoRA scaling α	16
LoRA dropout	0.05
Target modules	All linear layers in language backbone
Frozen modules	Vision encoder and multimodal projector
Optimizer	AdamW
Learning rate	5×10^{-5}
Batch size	1
Maximum epochs	10
LR scheduler	Linear decay
Warmup	None
Maximum gradient norm	1.0
Gradient checkpointing	Enabled
Precision	bfloat16
Early stopping patience	2 epochs
Minimum delta	10^{-4}
Early stopping start	After at least 3 epochs

Table 9: **Hyperparameters and multimodal adaptations for unlearning methods.**

Method	LR	Epochs	Batch	Key setting	Multimodal adaptation
GA	2×10^{-5}	1	1	Forget-set ascent	Native multimodal input
GA-Diff	2×10^{-5}	1	1	Forget ascent + retain descent	Native multimodal input
KL-Min	2×10^{-5}	1	1	Frozen Vanilla reference	Native multimodal input
NPO	2×10^{-5}	1	1	$\beta = 0.4$, retain-only reference	Native multimodal input
MMUN-Lang	1×10^{-5}	1	1	Language mask	Language-backbone updates
MMUN-Vis	1×10^{-5}	1	1	Vision mask	Vision-encoder updates
MMUN-Both	1×10^{-5}	1	1	Language + vision masks	Both components updated
R2MU	–	1	1	Original R2MU objective	Image–text QA flattened to text format
MANU	–	1	1	Original pruning procedure	Multimodal-format wrapper

adapt only the input interface. Multimodal image–text QA examples are flattened into the text-based sample format expected by the original implementation.

MANU. MANU performs unlearning by progressively pruning parameters associated with the forget data. We retain the original pruning procedure and introduce a wrapper that converts our multimodal supervision into the data format required by the original implementation.

The optimization objectives of GA, GA-Diff, KL-Min, and NPO operate directly on multimodal supervision without modifying their underlying formulation. For MMUN, R2MU, and MANU, the changes above concern parameter selection or input representation rather than introducing new unlearning objectives.

9.4 Hardware, Software, and Compute

All experiments are conducted on a server equipped with two NVIDIA A100 GPUs, each with 80 GB of memory. Table 10 summarizes the software and numerical configuration used throughout the experiments.

All models are run in bfloat16 precision with scaled dot-product attention enabled for computational efficiency.

10 Complete Quantitative Results

We report every protocol-verified model–method–setting combination. Missing or invalid runs should be marked explicitly rather than replaced by zeros.

10.1 Qwen3-VL Results under Forget-1 and Forget-5

Tables 11–13 report the complete Forget-1 and Forget-5 results for the three Qwen3-VL scales.

Table 10: **Hardware and software environment.**

Component	Configuration
GPU	2× NVIDIA A100 80 GB
CUDA	13.2
Python	3.10
PyTorch	2.4
Transformers	≥4.45
PEFT	≥0.12
Accelerate	≥0.34
Precision	bfloat16
Attention backend	SDPA

We report the four task families separately on both the forget and retain splits.

IB^+ is the recovery rate on matched same-identity pairs. It is reported as a behavioral binding diagnostic and should not be interpreted as balanced same/different classification accuracy. The Qwen3-VL-2B R2MU run under Forget-1 is omitted because its recorded split does not pass the benchmark alignment audit.

10.2 LLaVA-1.5 Results under Forget-1 and Forget-5

Tables 14 and 15 report the complete Forget-1 and Forget-5 results for LLaVA-1.5-7B and LLaVA-1.5-13B. We report the four task families separately on both the forget and retain splits.

LLaVA-specific bias on IB^+ . IB^+ evaluates recovery of matched identity bindings rather than same/different classification. All non-LLaVA Base models score zero, whereas LLaVA-1.5-7B answers *Yes* to all 480 probes and LLaVA-1.5-13B does so for 90.21% before profile fine-tuning. Since these identities are unseen, the high scores reflect a LLaVA-specific affirmative-response bias rather than acquired binding knowledge. We therefore report LLaVA IB^+ for completeness but exclude it from conclusions about binding removal.

10.3 Llama-3.2-11B Results under Forget-1 and Forget-5

Table 16 reports the complete Forget-1 and Forget-5 results for Llama-3.2-11B. We report the four task families separately on both the forget and retain splits.

11 Extended Analysis and Discussion

Masked evidence remains a substantial identity cue. Among Vanilla-correct queries, the average residual is 53.88% for direct identity ac-

cess and 51.85% for indirect identity access across the valid MMUN configurations. On Qwen3-VL-4B, MMUN-Lang retains 70.00% direct IA and 70.37% indirect IA. Removing a name or explicit identifier therefore changes the cue available to the model without necessarily removing the identity signal carried by the document layout and surrounding attributes.

Biometric continuity is more persistent than record linkage. Vanilla-conditioned recovery averages 94.00% for cross-instance binding and 81.80% for cross-modal binding over the valid MMUN configurations. All three MMUN variants preserve every eligible cross-instance binding on LLaVA-1.5-7B, LLaVA-1.5-13B, Qwen3-VL-2B, and Qwen3-VL-4B. On Qwen3-VL-8B, MMUN-Vis removes all eligible cross-modal bindings while 40.00% of cross-instance bindings remain recoverable. Since IB^+ contains matched positive pairs, these values measure persistence of stored bindings rather than balanced same/different classification.

Relational composition can recover an identity after direct access weakens. GA-Diff on Qwen3-VL-2B reduces forget AA to 44.76%, while forget IR remains 73.33%. MMUN-Lang on Qwen3-VL-4B shows the same separation, with 44.76% AA and 75.56% IR. More specifically, it retains every Vanilla-correct relational and relational-attribute reconstruction query on that model. Residual relations can therefore provide an access route even when an isolated attribute query has been suppressed.

Additional attributes do not guarantee stronger reconstruction. Across the MMUN configurations, Vanilla-conditioned residuals are 44.30% for relational reconstruction and 45.60% for relational-attribute reconstruction. The additional attribute can disambiguate a surviving relation, but it can also be unavailable after unlearning. Cross-modal

Table 11: Qwen3-VL-2B results under Forget-1 and Forget-5.

Method	Forget residual ↓				Retain utility ↑			
	AA	IA	IB ⁺	IR	AA	IA	IB ⁺	IR
<i>Forget-1</i>								
Base	16.67	27.27	0.00	66.67	13.96	26.89	1.27	56.69
Vanilla	100.00	63.64	25.00	88.89	99.88	68.64	34.96	74.95
GA	100.00	63.64	37.50	88.89	99.31	68.18	33.90	74.01
GA-Diff	64.29	59.09	37.50	88.89	83.01	63.64	55.30	76.84
KL-Min	100.00	59.09	50.00	88.89	99.27	68.49	43.64	74.20
NPO	92.86	63.64	37.50	88.89	99.72	68.72	37.29	76.08
MANU	100.00	63.64	50.00	88.89	99.88	68.80	34.11	75.33
MMUN-Both	23.81	0.00	75.00	11.11	61.34	54.01	40.25	64.97
MMUN-Lang	14.29	4.55	25.00	0.00	61.86	50.69	19.28	55.56
MMUN-Vis	14.29	18.18	75.00	66.67	61.14	52.70	74.36	61.39
<i>Forget-5</i>								
Base	15.24	27.27	0.00	27.78	13.90	26.86	1.36	26.77
Vanilla	98.57	68.18	42.50	77.78	100.00	68.60	34.09	74.95
GA	94.76	60.00	37.50	80.00	97.01	67.27	31.36	71.31
GA-Diff	44.76	55.45	77.50	73.33	73.25	57.52	59.09	68.28
KL-Min	98.10	68.18	47.50	80.00	99.78	69.50	40.91	74.55
NPO	85.24	66.36	40.00	75.56	97.01	68.43	35.00	70.91
MANU	98.57	68.18	45.00	75.56	100.00	68.76	33.41	75.56
MMUN-Both	41.90	32.73	85.00	48.89	78.61	62.23	74.32	57.58
MMUN-Lang	40.00	40.91	92.50	46.67	79.48	62.98	73.64	63.23
MMUN-Vis	32.38	36.36	70.00	55.56	76.54	59.67	54.32	55.56
R2MU	16.19	27.27	0.00	60.00	14.07	26.94	1.36	57.17

relational reconstruction has a lower Vanilla acquisition rate, ranging from 13.33% to 33.33% across backbones. Its raw residual should therefore be interpreted together with the number of Vanilla-correct examples.

Person–attribute edges are removed asymmetrically. For MMUN-Lang, attribute-to-identity recovery on Qwen3-VL-2B, 4B, and 8B is 66.67%, 100.00%, and 60.00%, whereas identity-to-attribute recovery is 16.67%, 40.00%, and 4.00%. On LLaVA-1.5-7B, reverse access is removed for every eligible pair while 20.00% retain only forward access. The surviving lookup direction depends on the backbone, so one query direction cannot certify deletion of the underlying association.

Textual and visual access can fail in opposite ways. On Qwen3-VL-2B, MMUN-Lang retains 24.64% of Vanilla-correct textual attribute queries and 47.83% of visual queries. On LLaVA-1.5-13B, MMUN-Vis retains 73.91% textual queries and 54.35% visual queries. The weaker modality path is therefore not fixed by the nominal modality of the unlearning objective. Shared multimodal representations can move residual access to another input route.

The effective modality target depends on the backbone. MMUN-Vis gives the lowest any-path recovery on LLaVA-1.5-7B and Qwen3-VL-2B, at 42.03% and 57.97%. It becomes the weakest MMUN variant on LLaVA-1.5-13B, where recovery reaches 82.61%. MMUN-Lang is strongest on LLaVA-1.5-13B, Qwen3-VL-4B, and Qwen3-VL-8B, with recovery rates of 65.22%, 66.67%, and 24.64%. A fixed language or vision mask does not identify a universal location of identity information.

Deletion load interacts with the backbone. For MMUN-Lang, increasing the forget set from one to five identities raises any-path recovery from 42.86% to 62.32% on LLaVA-1.5-7B, from 21.43% to 71.01% on Qwen3-VL-2B, and from 35.71% to 66.67% on Qwen3-VL-4B. The trend reverses on LLaVA-1.5-13B and Qwen3-VL-8B, where recovery falls from 92.86% to 65.22% and from 35.71% to 24.64%. Forget-1 is therefore a single-identity diagnostic, not a population-level estimate of multi-identity deletion.

Method rankings do not transfer reliably across backbones. GA leaves forget AA at 97.62% on LLaVA-1.5-7B, but reduces it to 54.76% on LLaVA-1.5-13B and 58.10% on Llama-3.2-11B.

Table 12: Qwen3-VL-4B results under Forget-1 and Forget-5.

Method	Forget residual ↓				Retain utility ↑			
	AA	IA	IB ⁺	IR	AA	IA	IB ⁺	IR
<i>Forget-1</i>								
Base	7.14	27.27	0.00	22.22	9.64	20.57	0.00	26.55
Vanilla	100.00	68.18	100.00	88.89	99.88	72.65	83.26	71.56
GA	100.00	68.18	100.00	88.89	99.88	72.65	83.05	71.00
GA-Diff	90.48	68.18	100.00	77.78	93.99	69.88	83.05	72.88
KL-Min	100.00	68.18	100.00	77.78	99.88	72.42	83.26	71.19
NPO	100.00	68.18	100.00	88.89	99.88	72.65	83.26	71.19
MANU	100.00	68.18	100.00	88.89	99.88	72.50	83.26	72.32
MMUN-Both	28.57	40.91	100.00	88.89	72.60	63.48	61.86	66.29
MMUN-Lang	23.81	9.09	0.00	11.11	68.64	61.09	58.90	63.09
MMUN-Vis	23.81	31.82	100.00	33.33	69.53	62.71	73.09	65.91
R2MU	9.52	27.27	0.00	22.22	9.36	21.11	0.00	26.74
<i>Forget-5</i>								
Base	10.48	20.91	0.00	24.44	9.52	20.66	0.00	26.67
Vanilla	98.57	70.00	100.00	71.11	100.00	72.81	82.05	71.92
GA	86.67	62.73	100.00	64.44	99.39	71.65	82.05	68.89
GA-Diff	80.00	68.18	100.00	77.78	88.66	69.59	81.82	76.97
KL-Min	98.57	70.00	100.00	71.11	100.00	72.98	81.82	72.12
NPO	97.14	70.00	100.00	68.89	99.96	72.56	82.05	72.32
MANU	98.57	70.00	100.00	71.11	100.00	72.64	82.05	72.73
MMUN-Both	57.62	61.82	100.00	75.56	87.75	69.09	85.68	69.09
MMUN-Lang	44.76	49.09	100.00	75.56	93.68	69.50	81.59	66.26
MMUN-Vis	50.48	55.45	100.00	48.89	89.78	68.93	81.36	65.86
R2MU	12.38	20.91	0.00	24.44	9.09	21.24	0.00	26.87

Their retain AA values are 66.45% and 90.48% for the latter two models. GA-Diff reaches 44.76% forget AA on Qwen3-VL-2B but 98.57% on LLaVA-1.5-7B. An unlearning objective can therefore appear ineffective, selective, or destructive depending on the backbone.

Alternative identity keys create disagreement rather than a single dominant path. Under MMUN-Lang on Qwen3-VL-2B, name, face, and fingerprint keys recover 24.64%, 55.07%, and 40.58% of eligible attributes individually, while their union recovers 71.01%. At 4B, the strongest individual path recovers 53.62%, but the union reaches 66.67%. At 8B, every individual path is below 16%, yet the union remains 24.64%. Different keys disagree on 20.29–56.52% of eligible cases. A low residual for one identity key does not establish that the attribute is inaccessible through another key.

The benchmark exposes complementary failure modes rather than one scalar notion of forgetting. Direct attribute access, identity access, binding, and reconstruction can move in different directions under the same update. A method can suppress a named answer while preserving a re-

verse lookup, a biometric link, or a relational reconstruction route. The full task profile is therefore necessary for interpreting whether an individual has been removed as a connected identity.

12 Limitations and Future Directions

Synthetic profiles and external validity. IDUnlearn-Bench uses synthetic individual profiles to provide controlled identity–attribute associations, cross-modal evidence, and reproducible forgetting targets without exposing real personal information. While this design enables precise construction and auditing of individual-level evidence, synthetic profiles cannot fully capture the diversity, noise, and ambiguity of naturally occurring personal data. The current benchmark therefore evaluates individual-level unlearning under a controlled setting rather than claiming to represent all real-world identity distributions. Future versions could extend the benchmark to broader linguistic, cultural, demographic, and temporal settings, as well as less structured evidence distributions.

Behavioral scope of the evaluation. Our evaluation measures whether target information remains

Table 13: Qwen3-VL-8B results under Forget-1 and Forget-5.

Method	Forget residual ↓				Retain utility ↑			
	AA	IA	IB ⁺	IR	AA	IA	IB ⁺	IR
<i>Forget-1</i>								
Base	7.14	13.64	0.00	22.22	7.71	16.02	0.00	29.19
Vanilla	100.00	68.18	37.50	77.78	99.88	72.96	53.60	73.63
GA	100.00	68.18	37.50	77.78	99.72	73.34	49.36	73.45
GA-Diff	78.57	59.09	12.50	77.78	92.62	69.18	44.49	74.39
KL-Min	100.00	68.18	37.50	77.78	99.72	72.73	51.69	73.07
NPO	100.00	68.18	62.50	77.78	99.84	72.96	52.97	73.82
MANU	100.00	68.18	50.00	77.78	99.88	72.88	53.18	73.63
MMUN-Both	28.57	4.55	12.50	11.11	68.04	61.63	33.47	69.11
MMUN-Lang	26.19	9.09	25.00	11.11	66.87	59.48	10.59	69.11
MMUN-Vis	23.81	4.55	62.50	11.11	64.53	61.25	12.29	67.61
R2MU	7.14	13.64	0.00	22.22	8.27	16.49	0.00	32.58
<i>Forget-5</i>								
Base	8.10	13.64	0.00	28.89	7.66	16.20	0.00	29.09
Vanilla	98.57	70.00	65.00	71.11	100.00	73.14	52.27	73.94
GA	91.90	69.09	65.00	73.33	99.39	72.07	49.09	73.33
GA-Diff	77.62	60.00	55.00	62.22	81.95	66.61	42.73	72.12
KL-Min	98.57	68.18	57.50	71.11	99.78	72.98	47.27	73.54
NPO	97.14	70.00	62.50	71.11	100.00	73.06	50.23	74.95
MANU	98.57	70.00	62.50	71.11	100.00	73.06	52.27	73.94
MMUN-Both	20.95	22.73	35.00	33.33	65.06	57.93	15.23	72.12
MMUN-Lang	14.29	20.00	77.50	33.33	64.03	58.76	30.23	69.90
MMUN-Vis	19.52	15.45	10.00	31.11	62.81	54.88	5.68	64.85
R2MU	8.57	13.64	0.00	35.56	8.23	16.69	0.00	32.12

behaviorally recoverable through the access, binding, and reconstruction paths instantiated by the benchmark. A low leakage score indicates that the evaluated queries fail to recover the corresponding information, but does not certify that all internal representations of the target have been removed from the model. As with other behavioral evaluations of machine unlearning, stronger adaptive probing or previously unseen access paths may reveal residual information not captured by the current task suite.

Toward stronger auditing and structure-aware unlearning. Future work should extend individual-level evaluation beyond fixed single-query tests toward adaptive multi-query recovery, multi-turn interaction, harder cross-modal association tests, and more diverse evidence graphs. More importantly, our results motivate unlearning methods that explicitly account for the structured nature of individual knowledge. Rather than removing isolated attributes alone, future methods could target identity–attribute associations, cross-modal links, and relational evidence while preserving information associated with unrelated individuals. We view such structure-aware unlearning as an important di-

rection exposed by IDUnlearn-Bench, rather than a capability assumed or implemented by the current benchmark.

Table 14: LLaVA-1.5-7B results under Forget-1 and Forget-5.

Method	Forget residual ↓				Retain utility ↑			
	AA	IA	IB ⁺	IR	AA	IA	IB ⁺	IR
<i>Forget-1</i>								
Base	14.29	27.27	100.00	55.56	10.17	23.42	100.00	52.73
Vanilla	100.00	68.18	100.00	66.67	99.64	68.34	83.05	71.19
GA	92.86	68.18	100.00	66.67	99.64	68.80	83.05	70.62
GA-Diff	100.00	59.09	100.00	77.78	99.60	69.11	83.05	70.62
KL-Min	100.00	68.18	100.00	66.67	99.64	68.26	83.05	70.24
NPO	100.00	68.18	100.00	66.67	99.64	68.49	83.05	71.00
MANU	100.00	68.18	100.00	66.67	99.64	68.72	83.05	70.62
MMUN-Both	16.67	9.09	87.50	22.22	95.00	65.56	85.17	66.48
MMUN-Lang	28.57	4.55	100.00	0.00	95.84	67.18	83.26	70.62
MMUN-Vis	23.81	9.09	87.50	0.00	94.92	68.26	83.05	68.55
<i>Forget-5</i>								
Base	13.33	29.09	100.00	62.22	9.96	22.98	100.00	51.92
Vanilla	98.57	70.00	100.00	75.56	99.74	68.18	81.82	70.71
GA	97.62	73.64	100.00	75.56	99.70	69.17	82.50	71.52
GA-Diff	98.57	47.27	100.00	66.67	99.74	67.36	81.82	69.70
KL-Min	98.57	70.00	100.00	75.56	99.74	68.26	81.82	70.30
NPO	98.57	57.27	100.00	77.78	99.74	67.77	81.82	70.10
MANU	98.57	67.27	97.50	75.56	99.39	65.04	69.32	69.70
MMUN-Both	33.33	19.09	100.00	33.33	94.59	66.45	85.45	70.10
MMUN-Lang	42.38	11.82	100.00	13.33	94.46	65.54	81.82	70.10
MMUN-Vis	25.71	9.09	100.00	13.33	93.90	65.79	81.82	67.07

Table 15: LLaVA-1.5-13B results under Forget-1 and Forget-5.

Method	Forget residual ↓				Retain utility ↑			
	AA	IA	IB ⁺	IR	AA	IA	IB ⁺	IR
<i>Forget-1</i>								
Base	11.90	31.82	75.00	66.67	9.04	25.19	90.47	54.61
Vanilla	100.00	68.18	100.00	77.78	99.60	68.34	83.05	58.76
GA-Diff	100.00	50.00	100.00	77.78	97.34	67.33	83.69	64.60
KL-Min	100.00	68.18	100.00	77.78	99.60	68.57	83.05	60.26
NPO	92.86	68.18	100.00	77.78	99.64	68.72	83.05	60.08
MANU	100.00	68.18	100.00	77.78	99.60	68.49	83.05	58.57
MMUN-Both	23.81	9.09	100.00	22.22	94.03	65.25	94.28	58.19
MMUN-Lang	69.05	18.18	100.00	11.11	98.47	65.95	86.86	65.73
MMUN-Vis	35.71	9.09	100.00	22.22	93.91	61.71	100.00	61.02
<i>Forget-5</i>								
Base	10.48	30.00	95.00	66.67	8.96	24.88	89.77	53.74
Vanilla	98.57	68.18	100.00	71.11	99.70	68.35	81.82	57.98
GA	54.76	68.18	100.00	53.33	66.45	66.78	100.00	50.71
GA-Diff	82.86	61.82	100.00	68.89	98.74	66.69	81.82	55.56
KL-Min	98.57	68.18	100.00	71.11	99.70	68.68	81.82	59.19
NPO	67.62	68.18	100.00	64.44	83.94	69.01	93.86	52.93
MANU	98.57	68.18	100.00	71.11	99.74	68.26	81.82	58.59
MMUN-Both	38.57	12.73	100.00	31.11	95.63	63.64	100.00	57.37
MMUN-Lang	42.38	21.82	100.00	31.11	94.55	62.48	81.82	54.34
MMUN-Vis	60.00	10.00	100.00	20.00	96.84	66.28	82.95	65.45

Table 16: **Llama-3.2-11B** results under **Forget-1** and **Forget-5**.

Method	Forget residual ↓				Retain utility ↑			
	AA	IA	IB ⁺	IR	AA	IA	IB ⁺	IR
<i>Forget-1</i>								
Base	0.00	9.09	0.00	33.33	4.12	12.17	0.00	22.03
Vanilla	100.00	59.09	0.00	88.89	99.88	62.48	0.64	81.54
GA	88.10	68.18	0.00	77.78	99.88	63.48	0.21	81.17
GA-Diff	100.00	59.09	0.00	88.89	99.39	60.86	0.00	78.34
KL-Min	100.00	59.09	0.00	88.89	99.88	61.86	0.00	80.41
NPO	100.00	59.09	0.00	77.78	99.88	62.10	0.42	81.36
MANU	100.00	59.09	0.00	88.89	99.72	60.55	0.64	79.10
R2MU	7.14	22.73	12.50	44.44	4.88	20.11	6.78	38.61
<i>Forget-5</i>								
Base	2.86	8.18	0.00	20.00	4.16	12.48	0.00	22.42
Vanilla	98.57	56.36	2.50	82.22	100.00	62.98	0.45	81.62
GA	58.10	74.55	0.00	77.78	90.48	63.80	0.00	77.98
GA-Diff	89.52	47.27	0.00	77.78	99.18	55.04	0.91	71.52
KL-Min	98.57	56.36	0.00	82.22	100.00	60.66	0.00	81.41
NPO	98.10	60.91	0.00	82.22	99.91	63.80	1.14	82.02
MANU	96.19	55.45	0.00	82.22	99.57	60.66	1.59	79.19
R2MU	8.10	20.91	5.00	28.89	4.63	20.08	7.05	39.60